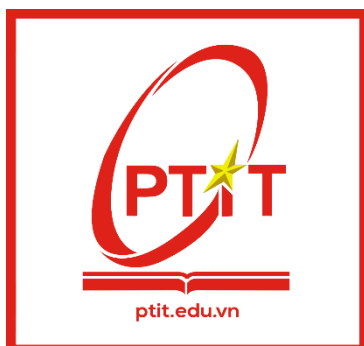


**POSTS AND TELECOMMUNICATIONS INSTITUTE OF
TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY**



**TERM PROJECT REPORT:
BUSINESS INTELLIGENCE**

Topic:

Social media data analysis for an automotive enterprise

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1. Problem requirements analysis: Social media data analysis for an automotive enterprise

The core objective of this component is to design and build a comprehensive social listening system that collects and analyzes domestic consumer discussions regarding VinFast's electric vehicles (EVs) and its competitors. The system transforms unstructured data into a Star Schema data warehouse and visualizes it via a self-developed Web Dashboard to ensure maximum smoothness and deep customization tailored to corporate needs.

1.1. Detailed System Objectives

- **Measuring Domestic Brand Health:** Quantify the level of interest and analyze the sentiment of Vietnamese customers toward VinFast's entire EV product lineup (from VF 3 to VF 9).
- **Tracking Market and Competitors:** Monitor user feedback on newly arrived foreign brands in Vietnam, such as BYD (Atto 3), Wuling (Mini EV), or Xiaomi (SU7), to benchmark VinFast's competitive advantages in charging infrastructure and after-sales service.
- **Instant Technical Crisis Warning:** Detect early negative comment streams related to software glitches, phantom current drain, and battery charging issues, enabling the engineering and PR departments to respond before a media crisis escalates.
- **Optimizing the Reporting Experience:** Instead of using out-of-the-box BI tools that face performance bottlenecks when handling large datasets, the enterprise will develop a proprietary Web Dashboard. This guarantees fluid data rendering, millisecond response times, and highly flexible analysis flow customization.

1.2. Data Ingestion Scope

The entire collection scope is limited to the Vietnam market and utilizes the Vietnamese language. The system extracts data from the following major platforms:

- **YouTube (Primary Supplementary Source):**
 - *Scope:* Collect data from car review videos on top-tier technology and automotive channels in Vietnam (e.g., Xế Cưng, Autodaily, Trắng Auto, Dương Đề, TipCar, Xe Tinh Tế...).
 - *Data Extracted:* Video identifiers, channel information, and the complete comment tree under these review videos.
- **Facebook Dataset:** Focus on discussion data from car owner communities (VF 3 Club, VinFast EV Vietnam Community, Trà đá ô tô...), review posts, and real-world experience sharing.
- **Reddit & Twitter (X):** Scan subreddits and hashtags where Vietnamese users discuss electric vehicles, such as r/VietNam and r/VinFast.

- **Tinh Te:** Extract articles and comments from the "Tinh Te Xe" section—a hub where tech-savvy consumers provide in-depth evaluations of vehicle software and operation.

1.3. Data Preprocessing Pipeline

To resolve the challenges of free-form Vietnamese text, raw data must pass through a strict processing pipeline:

- **Text Cleansing (Complete Emoji Removal):** Unlike standard NLP models that retain emojis to infer sentiment, this system requires the absolute removal (100%) of all symbols, emojis, and emoticons (e.g., 😡, 👍, 😂...). The goal is to eliminate maximum noise, reducing text to its pure string format to focus accurately on the user's semantics and syntax.
- **Vietnamese Tokenization & Processing:** Utilize specialized libraries for Vietnamese (such as Underthesea or VnCoreNLP) to perform compound word tokenization (e.g., "trạm_sạc", "lỗi_ảo", "thuê_pin").
- **Teencode and Slang Mapping:** Build an automated normalization dictionary to convert user abbreviations and slang back to standard orthography:
 - ko / k => không
 - đc / dc => được
 - ae => anh em
 - vinfet / vinfat => VinFast
- **Interrogative Sentence Filter:** Separate and isolate neutral investigative questions (e.g., "Xe này chạy có hay lỗi không mọi người?") from the Negative category, directing the AI model to label them as Neutral.

1.4. Data Analysis Requirements

1.4.1. Sentiment Analysis

The machine learning model will classify cleansed text into 3 distinct sentiment states: Positive, Neutral, and Negative, accompanied by a confidence score. The system must seamlessly handle complex Vietnamese sentence structures involving sentiment reversals (e.g., "Không hề lỗi" => Positive; "Cực kỳ thất vọng" => Negative).

1.4.2. Aspect-Based Sentiment Analysis

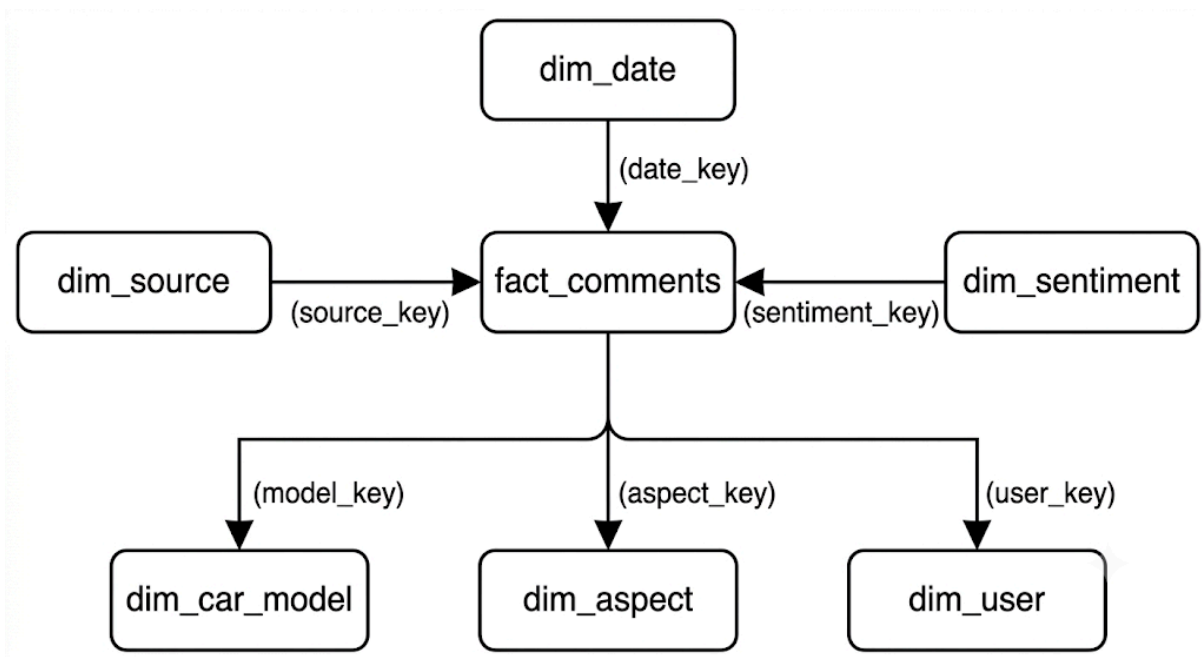
The system goes beyond general sentiment analysis to parse and accurately categorize text into 7 core aspects of business interest:

- **Charging Station (asp_charging):** Number of stations, charger distribution, charging speed, and situations where ICE (gasoline) cars occupy charging slots.
- **Battery & Battery Rental (asp_battery):** Battery rental policies, battery capacity, real-world driving range, and home charging.

- **Software & Minor Glitches (asp_software):** Screen freeze errors, ADAS phantom errors, and OTA software updates.
- **Operation & Comfort (asp_comfort):** Sound insulation, suspension performance, engine/motor responsiveness, air conditioning, and seat comfort.
- **Service & After-sales (asp_service):** Service center attitude, mobile rescue services, and waiting times for maintenance scheduling.
- **Pricing & Promotions (asp_price):** Promotional programs, on-road prices, and the application of Vinhomes vouchers.
- **Other (asp_other):** General discussions or topics falling outside the 6 categories above.

1.5. Data Warehouse Schema Specification

To ensure optimal performance for the Dashboard, the output data is stored centrally in a relational database format (compatible with the sentiment_dwh.db file) following a Star Schema with 1 Fact table and 6 closely linked Dimension tables:



Picture 1: Detailed Schema Structural Component Specifications:

- **Fact Table: fact_comments** (Stores comment/post records)
 - **comment_key** (TEXT - PK): Unique identifier of the comment (e.g., ytb_13597, gen_fa_818).
 - **date_key** (TEXT - FK): Links to the publication date.
 - **source_key** (TEXT - FK): Links to the collection channel and platform.
 - **sentiment_key** (TEXT - FK): Links to the sentiment analysis results.

- model_key (TEXT - FK): Links to the specific car model mentioned.
- aspect_key (TEXT - FK): Links to the specific content aspect of the comment.
- user_key (TEXT - FK): Links to the speaker's user profile.
- comment_text (TEXT): Original text content after being cleansed (completely stripped of emojis).
- like_count (INTEGER): Total engagement/likes for that comment.
- Car Model Dimension: dim_car_model
 - model_key (TEXT - PK): Car model code (e.g., model_vf3, model_byd_atto3).
 - model_name (TEXT): Commercial name of the vehicle (VF 3, VF 5, VF 8, VF 9, SU7, Mini EV...).
 - brand (TEXT): Manufacturer (VinFast, BYD, Wuling, Xiaomi, Other/General).
 - segment (TEXT): Vehicle segment (Mini-SUV, A-SUV, C-SUV, D-SUV, Sedan...).
- Aspect Dimension: dim_aspect
 - aspect_key (TEXT - PK): Aspect code (e.g., asp_charging, asp_software).
 - aspect_name (TEXT): Display name (Charging Station, Software & Minor Glitches, Battery & Battery Rental...).
 - description (TEXT): Detailed classification scope description for the aspect.
- Sentiment Dimension: dim_sentiment
 - sentiment_key (TEXT - PK): Sentiment code (e.g., sent_113).
 - sentiment_label (TEXT): Sentiment label (Positive, Negative, Neutral).
 - confidence_score (REAL): AI model's confidence weight for that specific label.
- Source Dimension: dim_source
 - source_key (TEXT - PK): Data source code (e.g., src_youtube_xe_cung).
 - channel_name (TEXT): Specific channel/group name (Xế Cưng, Tinh Tế Xe, VinFast Owner Club...).
 - platform (TEXT): Core platform (YouTube, Facebook, Reddit, Tinh Tế, Twitter).
- User Dimension: dim_user
 - user_key (TEXT - PK): Anonymized unique user ID.
 - username (TEXT): Account name or numerical identifier.

- author_type (TEXT): User categorization (Regular - Standard user, Celeb - Influencer/Car reviewer KOL).
- follower_count (INTEGER): Number of account followers (supports virality weight calculations).
- Date Dimension: dim_date
 - date_key (TEXT - PK), full_date (TEXT), day, month, year, quarter (INTEGER) to facilitate granular timeline breakdowns.

1.6. Custom Web Dashboard Requirements

To meet the standards of fluid data presentation, high rendering performance, and complete elimination of latency found in commercial BI tools, the system will be deployed via an independent, custom-built Web Dashboard. It will leverage optimized frontend technologies like React.js or Vue.js combined with Canvas/SVG charts from open-source libraries like ECharts or Chart.js.

The Web Dashboard layout is divided into 3 main visual sub-systems:

1.6.1. Brand Health & Trend Sub-system

- **Smooth Time-series Sentiment Chart:** Displays the fluctuations of Positive/Negative/Neutral discussion volumes over Days/Weeks/Months. Employs data lazy-loading techniques to ensure buttery-smooth timeline scrolling.
- **Engagement Metrics & Virality Speed Tracker:** Displays total likes and detects sudden discussion frequency spikes. The system automatically triggers a red alert on the web interface if negative technical error discussions exceed thresholds within a 1-hour window.
- **Share of Voice (SoV):** A pie or donut chart representing the percentage of Vietnamese users mentioning VinFast compared to its competitors (Xiaomi, BYD, Wuling).

1.6.2. Aspect-Based Sentiment Grid Sub-system

- **Heatmap Grid:** The vertical axis represents the 7 aspects (Charging Station, Software, Battery...), while the horizontal axis displays the vehicle models (VF 3, VF 5, VF 8, VF 9...). Each intersection visualizes the Positive/Negative ratio using varying color gradients. Management can see at a glance that *"VF 9 is receiving the most complaints regarding Software"* or *"VinFast's charging station network is getting a wave of Positive feedback."*
- **Trending Topic Clustering:** Displays the most frequently mentioned keywords using bar charts or a network graph. These keywords are completely stripped of emojis, preserving only the core terms (e.g., "lỗi_áo", "treo_màn_hình", "mát_sâu", "voucher").

1.6.3. Fact Comment Streamer Sub-system

- **Infinite Scroll Datagrid:** Allows VinFast business analysts to read the original text written by users directly (cleansed of emojis).
- **Dynamic Multi-condition Slicers:** Integrates fast-response filter buttons to instantaneously narrow down the comment list: Platform (YouTube Only) => Car Model (Select VF 5) => Aspect (Select Operation & Comfort) => Sentiment (Select Negative). The web page updates the list instantly without reloading, delivering an exceptionally fluid user experience.

2. Dataset Description

The dataset utilized in this study was automatically collected from major social media and video-sharing platforms. The research specifically focuses on reviews, discussions, and evaluations of automobile products currently circulating in the Vietnamese market. This primarily includes Vietnamese electric vehicles (VinFast VF3, VF5, VF6, VF7, VF8, VF9, VF e34) alongside direct competitors in the internal combustion engine (ICE) or hybrid segments from international brands (such as Mitsubishi X-Force, BYD Dolphin/M6, Wuling Bingo, Geely EX2, etc.). The dataset is structured and stored in CSV format to directly serve the consumer opinion-mining task.

2.1. Data Sources and Scale

- **Multi-Platform Data Sources:** Raw data was automatically extracted via the crawler.py script using public API connections (such as YouTube Data API v3) and data extraction libraries from four of the most popular platforms in Vietnam:
 - YouTube: Collected metadata and comments from leading and reputable automotive review channels in Vietnam, such as Tipcar TV, Dương Dê, Autovietnam, GearUpVN, Mê Xe, Autodaily.vn, Xem Xe Analytics, Xế Cộg, XE HAY...
 - Facebook: Extracted posts and discussion comments from major automotive forums and real-world owner communities, such as Oto+ (OtoFun), OFFB, VinFast Electric Vehicle Vietnam Club, VinFast EV Community, VinFast VF6 / VF7 / VF8 / VF9 Vietnam Club, Tinh tế Xe...
 - Reddit: Gathered multi-dimensional posts and discussion threads regarding the Vietnamese automotive market and manufacturers from subreddits like r/VietNam, r/VinFastComm (a specialized forum discussing VinFast), and r/cars...
 - Twitter/X: Collected status updates from official brand accounts or tech/EV experts (e.g., @VinFastOfficial), along with tracking

trending keywords and hashtags in the Vietnamese automotive market such as #VinFast, #VF6, #VF8, #OtoVietNam...

- **Detailed Dataset Scale (Before and After Preprocessing):** The data collection and normalization process yielded specific scales across the storage files as follows:
 - Video Statistics Dataset (youtube_video_stats.csv): Stores the catalog of review videos, product identifiers, and general engagement metrics, reaching a fixed size of 482 raw records (each record corresponds to the information of a specific car review video).
 - Raw Customer Comments Dataset (youtube_comments.csv): Aggregates all original user opinions and discussions collected from automotive review sources, totaling 21,191 raw records.
 - Processed Customer Comments Dataset (processed_comments.csv): To ensure high accuracy for sentiment analysis and topic modeling, the raw data underwent deep cleaning. After removing 1,851 inappropriate records, the final clean dataset contains exactly 19,340 quality-approved records.
 - Data exclusion criteria include: Extremely short comments lacking informational value (e.g., "ok", "châm", "f", "duyet"); comments in foreign languages outside the scope of this study (Thai, unrelated English); spam or promotional advertisements; and texts containing only noise such as special characters or excessive emojis. The noise-filtering rate accounts for approximately 8.73% of the initial raw comment volume.

2.2. Structure of Raw Information Fields

The detailed structure of the default data fields extracted from the platforms, along with their relational database linkages, is detailed below:

Table 1: Structure of the Video Statistics File (youtube_video_stats.csv)

This file stores the metadata of the review videos, managing the overview information of the media sources.

No.	Field Name	Data Type	Description	Real-World Example
1	video_id	String	Unique identifier of the video on YouTube (Primary Key).	i-rLUIZDCEA

2	title	String	Title of the car review video.	Đánh giá Vinfast VF6 - Người mua phải đấu tranh tư tưởng trước khi chốt TIPCAR TV
3	channel_title	String	Name of the YouTube channel that uploaded the video.	Tipcar TV
4	publish_date	Datetime	The time the video was published on the platform (ISO 8601).	2024-02-14T12:09:42Z
5	view_count	Integer	Total views accumulated at the time of data crawling.	314725
6	like_count	Integer	Total number of likes received by the video.	2915
7	comment_count	Integer	Total number of comments appearing under the video.	951

Table 2: Structure of the Raw Customer Comments File (youtube_comments.csv)

This file stores the detailed, unaltered feedback from the consumer community.

No.	Field Name	Data Type	Description	Real-World Example
1	video_id	String	Identifier of the video containing the comment (Foreign Key).	i-rLUIZDCEA

2	author	String	Username or anonymous identifier of the commenter.	@TuyenNguyen-xq4xc
3	text	String	Raw comment text typed by the user (containing praise, criticism regarding price, performance, battery, etc.).	"Cái ăn tiền nhất của xe này là sử dụng tiết kiệm thì kênh ko nói Tầm 500đ/1km Review quá thiếu sót"
4	like_count	Integer	Number of likes this specific comment received from other viewers.	118
5	published_at	Datetime	The timestamp when the user submitted the comment (ISO 8601).	2024-02-14T13:46:37Z

Table 3: Structured Attributes Post-Preprocessing (processed_comments.csv)

The final output file inherits the entire structure of the raw data table but introduces a core field processed via Natural Language Processing (NLP):

Additional Field Name	Data Type	Description	Normalized Data Example
processed_comment	List / String	Text content that has been lowercased, de-noised, tokenized, and stripped of stopwords.	['cái', 'ăn tiền', 'nhất', 'của', 'xe', 'này', 'là', 'sử dụng', 'tiết kiệm', 'thì', 'kênh', 'không', 'nói', 'tầm', '500', 'đ', 'l', 'km', 'review', 'quá', 'thiếu sót']

Data Linkage: The system manages these datasets using a strict one-to-many relationship. A single video review record in youtube_video_stats.csv directly maps to multiple user comments in youtube_comments.csv and processed_comments.csv via the shared video_id key.

2.3. Specialized Text Preprocessing Pipeline

Since data collected from social media and video-sharing platforms consists of unstructured User-Generated Content (UGC), it inherently contains a high noise ratio. To construct the clean processed_comments.csv dataset, a rigorous five-step Vietnamese Natural Language Processing (NLP) pipeline was designed and implemented:

[Raw Data] → (1) Lowercasing & Emoji Normalization → (2) Teencode & Abbreviation Mapping

→ (3) Vietnamese Tokenization → (4) Stopword & Noise Filtering →

[Clean Data]

- Text Normalization: All text strings are converted to lowercase to eliminate case variance. Non-semantic special characters (e.g., ~, *, @, #...) are stripped, and emoticons/emojis are either removed or mapped to textual representations depending on the downstream sentiment analysis requirements.
- Abbreviation and Teencode Resolution: This is a critical step for handling Vietnamese social media linguistics. The system utilizes a specialized mapping dictionary to standardize slang, jargon, and shorthand frequently used by consumers in automotive discussions:
 - ko, k => không (no/not)
 - đ, k (when denoting currency) => đồng, nghìn (e.g., 500đ => 500 đồng)
 - review, dì viuu => đánh giá (review/evaluate)
- Vietnamese Tokenization: Unlike English, where words are naturally separated by spaces, Vietnamese utilizes compound words (e.g., "ăn tiền" - value-for-money, "tiết kiệm" - economical, "giảm tốc" - decelerate). The pipeline integrates specialized Vietnamese tokenization libraries (such as Underthesea or VnCoreNLP) to accurately identify and bind these compound words into single tokens (e.g., ['ăn tiền', 'tiết kiệm', 'giảm tốc']) rather than splitting them into meaningless individual syllables.
- Stopword and Non-semantic Token Filtering: High-frequency words that lack semantic weight or sentiment expression (e.g., ạ, dạ, nhé, thì, mà, là...) are systematically filtered out to optimize model performance.
- Quality Threshold Filtering: Comments falling below a minimum length threshold or identified as automated system spam are completely discarded. As detailed in the dataset scale section, this step successfully eliminated 1,851 noisy records.

2.4. Technical Challenges in Data Collection & Solutions

Documenting engineering bottlenecks and their respective workarounds demonstrates technical competency and system design maturity:

- **Challenge 1: API Rate Limits and Quota Constraints:** The YouTube Data API v3 enforces a strict daily limit of 10,000 quota units. When querying an extensive matrix of keywords (from VF3 to VF9, BYD, Wuling, etc.) simultaneously, the application risks exhausting its quota or encountering IP rate-limiting blocks from public endpoints on Reddit and Facebook.
 - **Solution:** The crawler.py script incorporates an automated adaptive delay mechanism (`time.sleep(1)`) between consecutive API calls. Furthermore, the `maxResults` parameters are programmatically optimized to retrieve only high-relevance payloads, significantly conserving bandwidth and API quota allocations.
- **Challenge 2: Hidden Data and Structural Asymmetry:** Certain high-traffic videos from prominent KOLs have comment sections disabled, and various Facebook posts are protected by restrictive privacy settings, triggering unexpected exceptions (`HttpError`) during execution.
 - **Solution:** The ingestion logic is entirely wrapped within robust exception-handling blocks (`try-except`). When encountering access restrictions or system anomalies, the script logs a warning, gracefully bypasses the faulty record, and transitions to the next iteration without causing terminal crashes or data corruption.

2.5. Data Ethics & Privacy Compliance

Including a data ethics section is highly regarded in modern academic research and corporate-grade projects, ensuring compliance with international data governance standards.

This research strictly adheres to global data ethics standards and applicable cybersecurity regulations:

- **Platform Terms of Service Compliance:** The data mining process fully complies with the YouTube Terms of Service and the public data access policies of the respective platforms. The collected data is strictly designated for non-profit, academic research purposes.
- **User Anonymization:** Although the raw dataset captures the visible user display name (author), all personally identifiable information (PII) is strictly anonymized, hashed, or omitted during the downstream analysis and final reporting phases. Only the textual content (`text / processed_comment`) is retained for aggregate trend analysis. The system does not store user avatars, unique profile IDs, or any direct hyperlinks pointing to individual personal profiles.

2.6. Downstream Data Applications

This section outlines the strategic positioning of the cleaned dataset, paving the way for the methodology and empirical analysis chapters:

Leveraging the rigid one-to-many relational architecture established between the datasets, this data asset is positioned to resolve three core downstream analytical tasks:

1. **Consumer Sentiment Analysis:** Classifying user feedback into three discrete granularities: Positive (praising design, cost efficiency), Negative (criticizing stiff suspension systems, software glitches), and Neutral (inquiring about pricing or technical specifications).
2. **Topic Modeling (via LDA):** Automatically clustering customer discussions to uncover the most pressing dimensions driving the Vietnamese automotive market (e.g., charging infrastructure availability, on-road pricing, vehicle reliability).
3. **Media Engagement & Influence Analysis:** Cross-referencing engagement metrics from `youtube_video_stats.csv` (views, likes) with the extracted sentiment profiles from `processed_comments.csv` to measure how heavily tech KOLs sway public consumer sentiment in Vietnam.

3. Data Warehouse Design

3.1. Architectural Overview & Design Paradigm

The Data Warehouse (DWH) for the VinFast Social Listening & Sentiment Analysis System is explicitly engineered using a Star Schema design paradigm. This architectural choice is specifically optimized for Online Analytical Processing (OLAP) workloads, where read-heavy, multi-dimensional, and ad-hoc analytical queries predominate over transactional updates. The system structures historical data into a single, high-density central Fact Table surrounded by six highly descriptive Dimension Tables.

Compared to highly normalized database structures (such as 3NF) or Snowflake configurations, the Star Schema delivers profound technical and performance advantages within an enterprise data pipeline:

- **Minimized Computational Complexity:** Slicing and dicing millions of social media interactions requires minimal SQL JOIN operations. Queries only navigate a single layer of relationships from the central fact table out to the dimensions, vastly accelerating execution times and lowering memory overhead.
- **Business Intelligence (BI) Engine Readiness:** Leading enterprise BI and visualization applications—such as Power BI, Tableau, and Streamlit—are natively optimized to auto-detect and map relationships organized in a star format. This simplifies the creation of semantic layers, calculated measures, and rich interactive components.

- Flexible Multi-Dimensional Slicing: Business analysts can effortlessly cross-reference quantitative engagement indicators (such as like counts) across diverse qualitative categories, including temporal trends, sentiment variations, specific car models, user segments, and technical aspect failure points.

3.2. Central Fact Table Specification (fact_comments)

The central fact table, fact_comments, serves as the operational core of the data warehouse. It records every individual user comment event harvested from tracked digital environments. It maintains transactional granularity at the individual comment level, capturing foreign keys (FK) that route back to the surrounding dimensions, a text field for the processed message content, and quantitative metrics (measures) used for mathematical aggregations.

Column Name	Data Type	Key Type	Functional Description
comment_key	VARCHAR(100)	PK	The unique system identifier for each harvested comment event.
date_key	VARCHAR(50)	FK	Relational bridge pointing to the time dimension table (dim_date).
source_key	VARCHAR(50)	FK	Relational bridge pointing to the discussion source table (dim_source).
sentiment_key	VARCHAR(50)	FK	Relational bridge pointing to the sentiment category table (dim_sentiment).
model_key	VARCHAR(50)	FK	Relational bridge pointing to the vehicle product catalog (dim_car_model).

Column Name	Data Type	Key Type	Functional Description
aspect_key	VARCHAR(50)	FK	Relational bridge pointing to the granular technical aspect index (dim_aspect).
user_key	VARCHAR(50)	FK	Relational bridge pointing to the author profile dimension (dim_user).
comment_text	TEXT	Attribute	The alphanumeric text body of the comment (useful for deep NLP search or UI rendering).
like_count	INTEGER	Measure	An additive numerical indicator tracking total likes, used to weigh customer impact and message amplification.

3.3. Multi-Dimensional Schema Dictionary (Dimension Tables)

Dimension tables store textual descriptions, hierarchies, and qualitative parameters that add contextual depth to the facts. The following sub-sections provide the explicit database dictionary for each satellite table.

3.3.1. Time Dimension Table (dim_date)

The dim_date dimension decomposes raw timestamps into granular, pre-calculated temporal units. This removes the need for expensive datetime function calls during runtime query parsing and fully empowers long-term timeline charting, seasonality checking, and quarterly tracking.

Column Name	Data Type	Functional Mapping Description
date_key	VARCHAR(50)	Primary Key; formatted string acting as a smart key index (e.g., '20260616').
full_date	DATE	The complete standardized SQL date entry format (YYYY-MM-DD).
day	INTEGER	The precise numerical calendar day value (integer limits between 1 and 31).
month	INTEGER	The numerical calendar month value (integer limits between 1 and 12).
year	INTEGER	The full calendar year representation value (e.g., 2024, 2025, or 2026).
quarter	INTEGER	The business calendar quarter sequence value (ranges between 1 and 4).

3.3.2. Discussion Source Dimension Table (dim_source)

The dim_source dimension encapsulates metadata regarding original social environments, allowing cross-platform comparisons of where consumer public opinions originate.

Column Name	Data Type	Functional Mapping Description
source_key	VARCHAR(50)	Primary Key; unique hash identifier assigned to a specific media channel ecosystem.
channel_name	VARCHAR(255)	Specific community node or account publisher (e.g., Tipcar, Xe Hay, Autodaily, Tinh Tế).
platform	VARCHAR(50)	The master social media ecosystem hosting the event (YouTube, Facebook, Reddit, Twitter, Tinh Tế).

3.3.3. Sentiment Classification Dimension Table (dim_sentiment)

The dim_sentiment dimension isolates qualitative outputs assigned by the pipeline's natural language understanding algorithms (hybrid PhoBERT deep learning pipeline and fallback rule engines), alongside structural measurement confidence metrics.

Column Name	Data Type	Functional Mapping Description
sentiment_key	VARCHAR(50)	Primary Key; unique system reference code assigned to each unique sentiment combination.
sentiment_label	VARCHAR(50)	The classified emotional tone of the statement text (Positive, Negative, Neutral).
confidence_score	REAL / FLOAT	Model prediction accuracy probability float score value ranging precisely between 0.0 and 1.0.

3.3.4. Car Model Product Catalog Dimension (dim_car_model)

The dim_car_model dimension maps text statements directly onto distinct products. This lets the business track and evaluate brand share of voice, segment popularity, and competitive threats.

Column Name	Data Type	Functional Mapping Description
model_key	VARCHAR(50)	Primary Key; unique system reference identifier assigned to each automotive model entry.
model_name	VARCHAR(100)	The commercial moniker of the target vehicle (VF 3, VF 5, VF e34, VF 6, VF 7, VF 8, VF 9).
brand	VARCHAR(100)	The automotive manufacturer brand (VinFast or foreign market market rivals: BYD, Wuling, Xiaomi).
segment	VARCHAR(50)	Market size categorization classification segment (Mini EV, A-SUV, B-SUV, C-SUV, D-SUV, E-SUV).

3.3.5. Technical Aspect Dimension Table (dim_aspect)

The dim_aspect table maps statements onto operational, technical, or commercial components. This acts as an early warning indicator for quality control, pinpointing exact areas of product friction or failure.

Column Name	Data Type	Functional Mapping Description
aspect_key	VARCHAR(50)	Primary Key; unique system reference code assigned to each targeted technical/commercial aspect.
aspect_name	VARCHAR(100)	Grouped focus classification name (Charging Station, Battery & Leasing, Software, Operation, Service, Price).
description	TEXT	Comprehensive text scope boundary limits detailing matching keywords utilized by Regex/Mapping parsers.

3.3.6. User & Author Demographics Dimension (dim_user)

The dim_user table records metadata about authors, helping analysts track message amplification, reach, and separate comments by key industry influencers versus standard users.

Column Name	Data Type	Functional Mapping Description
user_key	VARCHAR(50)	Primary Key; unique system reference hash tracking specific social media user handles.
username	VARCHAR(100)	The profile name or numerical account identifier string of the comment author.
author_type	VARCHAR(50)	Author classification group type (Celeb/KOL or Regular User profiles).
follower_count	INTEGER	The total count size of followers, serving as a key metric to weigh message reach.

3.4. Relational Dynamics & Structural Enforcement

To guarantee strict analytical precision and data quality across downstream dashboards, the database layout applies rigid relational constraints based on the architectural design:

- One-to-Many Primary Cardinality (1:N): Each satellite dimension table maintains a strict One-to-Many relationship with the central fact_comments table. A precise product entry (e.g., 'VF 3'), calendar date, or specific user

profile can be linked to thousands of individual comment entries within the central fact table.

- **Foreign Key Enforcement Rules:** The ETL data loading pipeline strictly enforces referential integrity. Any incoming record processed into the fact_comments table must reference a key that already exists in the corresponding dimension table. If a query attempts to load a comment mentioning a brand-new vehicle model or an unindexed date, the pipeline throws an alert or dynamically provisions the key using INSERT OR IGNORE/upsert strategies to completely prevent orphan data fragments.

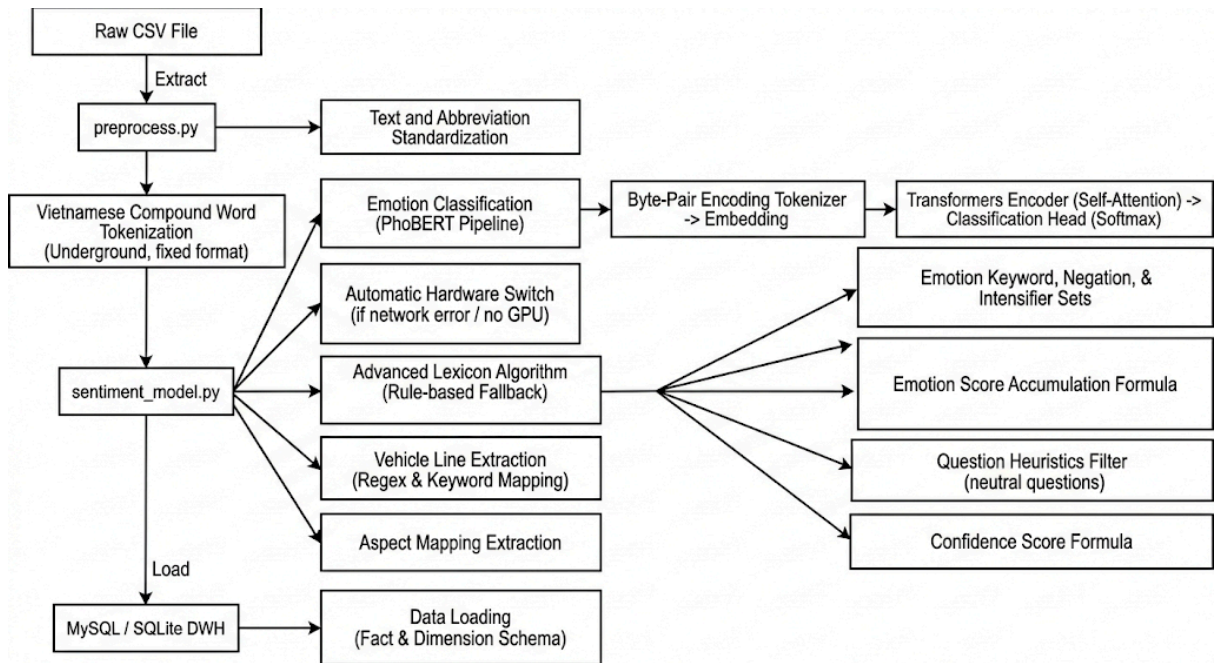
3.5. Enterprise Scalability & Operational Evaluation

The chosen star schema configuration delivers exceptional operational flexibility, supporting long-term technical modifications and system expansions:

1. **Effortless Aspect and Product Catalog Expansion:** As VinFast engineering groups roll out future vehicle platforms or track new commercial issues (e.g., autonomous driving functions or domestic solar charging networks), data engineering teams can simply append new records directly into dim_car_model or dim_aspect. There is zero need to rewrite complex code or run database structural alterations (ALTER TABLE) against the heavy central fact table.
2. **Real-Time Streaming Adaptability:** The warehouse's dual-mode strategy—using MySQL for live corporate deployment and SQLite for offline sandboxing—ensures high portability. This structure can seamlessly scale up to ingest live streaming message lines from streaming clusters like Apache Kafka or Spark Streaming, gracefully handling massive data influxes during major corporate PR events or product launches.

4. ETL Process (Extract - Transform - Load)

The ETL process is responsible for handling the entire data lifecycle, from raw data collection to structured storage in a data warehouse.



Picture 2: ETL Process

Raw social media data is very noisy. The preprocessing process is built through three rigorous steps:

4.1. Preprocess Phase (Transform)

4.1.1. Text Cleaning:

- **Unicode Normalization:** Convert the entire text to the NFC built-in standard (using `unicodedata.normalize('NFC')`) to eliminate font errors caused by built-in or combined Vietnamese characters.
- **Lowercase Conversion:** The entire string is converted to lowercase for vocabulary synchronization.
- **Noise Removal:** Use regular expressions (Regex) to remove URLs (regex `https?://[S+|www\.S+)`), email addresses (`\S+@\S+`), tags, special characters, emojis, and retain Vietnamese letters, numbers, and spaces.
- **Whitespace Normalization:** Remove extra leading/trailing spaces and reduce multiple spaces between words to a single space.

4.1.2. Standardization of slang and abbreviations (Slang Translation):

The system searches comments using a dictionary mapping common abbreviations and slang used on Vietnamese social media (automatically decomposed for AI model understanding):

- ko, k, khg, kg → khong
- dc, dc → duoc
- vfs, vf → vinfast
- e34, vfe34 --> vinfast vfe34

- ok, oke, okay --> good (for Lexicon to calculate positive points)
- ae --> anh em, nv --> nhan vien, tks/thanks --> thank you

4.1.3. Vietnamese Word Segmentation:

Vietnamese is a monosyllabic language, where word boundaries do not coincide with whitespace boundaries (for example, "xe điện" is a two-syllable compound word).

- The system uses the Underthesea library with the format="fixed" option to perform word segmentation.
- The output will automatically connect the compound words using underscores (for example: "trạm tích" →tram_sạc, "cảm giác lái" →cảm_giác_lái).
- This step is crucial for the PhoBERT model to correctly identify the token and prevent Lexicon from misinterpreting its meaning (for example, "tiết kiệm" is treated as a single token tiết_kiệm, avoiding independent evaluation as "tiết" and "kiệm").

4.2. Processing Mechanism of the Deep Learning Model (PhoBERT Sentiment Model)

The PhoBERT integrated model system (w11wo/phobert-base-vietnamese-sentiment) is a self-learning machine learning model (Self-learning Transformer) specifically for Vietnamese. The processing method is as follows:

1. Tokenization:

- Using the Byte-Pair Encoding (BPE) algorithm trained on PhoBERT's massive Vietnamese corpus, sentences are broken down into subword tokens.
- For example, the sentence: "VinFast electric car is very good" will be converted into corresponding integer ID tokens in PhoBERT's dictionary.

2. Embedding:

- The ID tokens are passed through the Embedding layer to be converted into multi-dimensional feature vectors (usually 768 dimensions).
- This Embedding layer integrates a Position Embedding mechanism so that the model understands the order of words in the sentence.

3. Contextual Encoding using Self-Attention:

- The core of PhoBERT is the Transformer Encoder architecture. The Multi-Head Self-Attention mechanism allows the model to calculate the contextual relationship between every word and all the other words in the sentence.

- Therefore, the word "error" in the sentence "software is error-free" will strongly focus on the preceding negative word "no," helping the model understand the actual semantic meaning as Positive instead of being mislabeled.

4. Emotion Classification (Classification Head):

- The final contextual vector of the special token [CLS] (representing the entire sentence) is passed through a Linear Layer to transform it into 3 logit values corresponding to 3 classes: Negative (0), Neutral (1), Positive (2).
- These logits are passed through the Softmax activation function to convert them into a probability distribution P for each class c :
- $P(y = c | x) = \exp(z_c) / (\exp(z_0) + \exp(z_1) + \exp(z_2))$

Where: z_c is the logit of class c , and x is the input sequence.

- The label with the highest probability will be selected as the predicted sentiment label, and the corresponding probability value is stored as the Confidence Score.

4.3. Advanced Lexicon Algorithm & Calculation Formula (Rules-based Fallback)

When the system automatically switches to the advanced Lexicon fallback mechanism, the sentiment and reliability assessment is calculated according to the following steps and logical expressions:

4.3.1. Defining the Vocabulary Dataset

- Pos: consists of keywords with positive connotations (beautiful, good, comfortable, cheap, enjoyable, solid, etc.).
- Neg: consists of keywords with negative connotations (faulty, boring, poor, expensive, terrible, broken, noisy, bumpy, etc.).
- Negotiation: consists of negative words ("not", "not yet", "nothing", "not at all", "cannot").
- Intensifier: consists of words that enhance sentiment ("very", "too", "much", "extremely", "very", "quite").

4.3.2. Cumulative Emotional Score Calculation Formula

For each comment containing a list of separated words $T = \{t_1, t_2, \dots, t_N\}$, the positive score (Score_pos) and negative score (Score_neg) are calculated cumulatively as follows:

- $\text{Score_pos} = \text{Sum of } [\text{Intensity}(t_i) * \text{PolaritySwap}(t_i, \text{Positive})]$ for every word t_i in the sentence.
- $\text{Score_neg} = \text{Sum of } [\text{Intensity}(t_i) * \text{PolaritySwap}(t_i, \text{Negative})]$ for every word t_i in the sentence.

Where:

- **Emotional Intensity (Intensity(t_i)):** Consider the window size of the two words preceding the current word t_i , i.e., $\{t_{i-2}, t_{i-1}\}$:
 1. Intensity(t_i) = 1.5 if there is a word from the Intensifier set in the two preceding words.
 2. Intensity(t_i) = 1.0 for all other cases.
- **The negative polarity reversal function (PolaritySwap(t_i , Type)):** Considers the appearance of a negative word in the window of the two preceding words:
 1. HasNegation(t_i) = 1 if there is a word from the Negation set in the two preceding words.
 2. HasNegation(t_i) = 0 otherwise.

When a negation is detected (HasNegation = 1), the word's polarity will be reversed:

- For a positive word (t_i belongs to Pos):
 1. PolaritySwap(t_i , Positive) = 0 if HasNegation(t_i) = 1 (the positive word is negated to negative).
 2. PolaritySwap(t_i , Positive) = 1 if HasNegation(t_i) = 0.
 3. PolaritySwap(t_i , Negative) = 1 if HasNegation(t_i) = 1.
 4. PolaritySwap(t_i , Negative) = 0 if HasNegation(t_i) = 0.
- For the negative word (t_i belongs to Neg):
 1. PolaritySwap(t_i , Negative) = 0 if HasNegation(t_i) = 1 (the negative word is negated to positive, e.g., "no error").
 2. PolaritySwap(t_i , Negative) = 1 if HasNegation(t_i) = 0.
 3. PolaritySwap(t_i , Positive) = 1 if HasNegation(t_i) = 1.
 4. PolaritySwap(t_i , Positive) = 0 if HasNegation(t_i) = 0.

4.3.3. Neutral Question Heuristics Filter

To avoid mistakenly classifying user information search questions as negative, the system applies a logical filtering condition:

- Set the logical function IsQuestion(text) = True if the sentence ends with a question mark (?) or contains question words from the QuestionMarkers set (e.g., "no everyone", "is ... not", "why", "is that okay").
- If IsQuestion(text) = True and the emotional difference is small:
 $Absolute_value(Score_pos - Score_neg) \leq 1.0$

The system will automatically assign the emotional label to **Neutral**.

4.3.4. Final Decision & Confidence Score

- **Case 1 (Neutral due to question or score balance):** If the question satisfies the condition IsQuestion(text) = True above, or if $Score_pos == Score_neg$:
Label = "Neutral", Confidence = 0.5
- **Case 2 (Positive):** If $Score_pos > Score_neg$:

Label = "Positive", Confidence = $\text{Min}(0.99, 0.6 + (\text{Score_pos} - \text{Score_neg}) * 0.1)$

- **Case 3 (Negative): If $\text{Score_neg} > \text{Score_pos}$:**

Label = "Negative", Confidence = $\text{Min}(0.99, 0.6 + (\text{Score_neg} - \text{Score_pos}) * 0.1)$

4.4. Aspect & Car Model Mapping

After preprocessing and sentiment analysis, the system extracts entity information using contextual keyword mapping:

- **Extracting Car Models:** Applying case-insensitive Regex matching to map car keywords to their corresponding Dim keys. Example:
 - vf3, vf 3 $\Rightarrow$ model_vf3 (VF 3 series)
 - vf e34, vfe34, e34 $\Rightarrow$ model_vfe34 (VF e34 series)
 - atto 3, atto3, byd $\Rightarrow$ model_byd_atto3 (BYD Atto 3 series)
- **Extracting Discussion Aspect:** Analyze the appearance of characteristic keywords in the sentence to classify them into 6 analysis dimensions:
 - Keywords: charging, charging_station, charging_pole $\rightarrow$ asp_charging (Charging station).
 - Keywords: battery, battery_rental, bms $\rightarrow$ asp_battery (Battery & Battery Rental).
 - Keywords: software, adas, minor_bugs, sensor, virtual_bugs $\rightarrow$ asp_software (Software & Minor Bugs).
 - Keywords: driving, firm, noisy, smooth, suspension, seat, interior $\rightarrow$ asp_comfort (Operation & Comfort).

4.5. Data Loading Phase

Database supports flexible dual mode:

- **MySQL (Online):** The official DWH connects directly via *pymysql* to update business reports.
- **SQLite (Offline):** The database file *sentiment_dwh.db* operates independently, requiring no server installation.
- The data writing process uses transaction structures to ensure data integrity: erasing old data in the Fact and Dimensions tables using TRUNCATE (MySQL) or DELETE (SQLite) before loading a new batch of data, ensuring no duplication when rerunning the test pipeline. The fact_comments event table loading phase strictly checks foreign key constraints with dimension tables.

5. Dashboard Development

The dashboard was developed using Streamlit combined with the Plotly graphics library to provide the smoothest and most intuitive BI analysis experience.

5.1. Interface Design & Aesthetics

- **Main color tone:** Royal Blue (#00529C), representing the VinFast brand color, combined with the dark interface of the Sidebar (#0b1a30) creates a professional and modern feel.
- **Typography & Font:** Integrated Google Font Outfit with a minimalist, easy-to-read style.
- **Animation:** Rounded chart corners, applied box-shadow effects, and smooth animation when hovering over Metric tabs or charts.
- **Contrast:** Completely resolved the issue of dark text on a dark background by forcing all text displayed on the Sidebar to bright white (#ffffff and #e2e8f0).

5.2. Details of the functions on the Sidebar (Left Toolbar)

The Sidebar is designed to act as a central control panel, allowing users to customize data connections, filter dynamic reports, and extract data. Detailed functions include:

5.2.1.Database Connection Manager (Database Configuration):

- Database Type (Selectbox): Allows quick switching between SQLite (Offline) (default) and MySQL (Online) data sources.
- MySQL Parameter Fields (Dynamic): When selecting MySQL (Online) connection, the Sidebar will open interactive input fields:
 - MySQL Host: Enter the IP address or MySQL server name (default localhost).
 - MySQL Port: Enter the connection port (default 3306).
 - MySQL User: Enter the MySQL account (default root).
 - MySQL Password: Password input type.
 - MySQL Database: Target schema database name (default sentiment_dwh).

5.2.2.Multi-Dimensional Report Filter (Report Filter):

This filter uses a dynamic multi-dimensional linking mechanism. When a filter changes, the related data in other filters and the entire chart on the Main Panel will automatically recalculate instantly:

- **Brand (Multiselect):** Filter data by car brand (VinFast, BYD, Wuling, Xiaomi).
- **Car Model (Multiselect):** Automatically change the displayed list based on the brands selected in the filter above (for example, if only VinFast is

selected, the list of car models will only show VF 3, VF 5, VF e34, VF 6, VF 7, VF 8, VF 9).

- **Feedback Aspect (Multiselect):** Filter by 6 technical/commercial aspects (Charging Stations, Battery & Battery Rental, Software & Troubleshooting, Operation & Amenities, Service & After-Sales Service, Price & Promotions).
- **Data Source (Multiselect):** Filter by social media platform (YouTube, Facebook, Twitter, Reddit, Tinh Te).
- **Emotion (Multiselect):** Quickly filter by sentiment label (Positive, Negative, Neutral).
- **Author Group (Multiselect):** Filter by author credibility level (Celeb/KOLs vs Regular Users).
- **Follower Count (Range Slider):** A slider that moves from 0 to the highest number of followers in the dataset to analyze the desired reach.

5.2.3.Data Export System for BI Tools (Export Data for BI Tool):

The sidebar provides two file export options corresponding to the storage architecture of Business Intelligence tools:

- **Data Export Format (Selectbox):** Allows selection between:
 - **Flat Table:** Suitable for quick loading into Tableau without needing table links. The system provides a **Download Flat CSV Table** button to download the filtered `vinfast_sentiment_flat.csv` file.
 - **Star Schema:** Suitable for building professional entity relationships in Power BI. When this mode is selected, the Sidebar will display 7 independent CSV download buttons corresponding to the DWH structure:
 1. **Download Fact_Comments.csv:** Downloads the facts table containing comments, likes, and associated foreign key fields.
 2. **Download Dim_User.csv:** Downloads the author information table, author type, and followers.
 3. **Download Dim_Source.csv:** Downloads the list of channels and platforms.
 4. **Download Dim_Date.csv:** Downloads the hierarchical time dimension table (Day, Month, Year, Quarter).
 5. **Download Dim_Sentiment.csv:** Downloads the sentiment label and trust score.
 6. **Download Dim_Car_Model.csv:** Download a list of car models, brands, and electric vehicle segments.
 7. **Download Dim_Aspect.csv:** Download a list of car aspects and their descriptions.

5.3. Components on the Main Panel

The main panel is divided into visual display areas from overview to detail:

5.3.1.KPI Metrics Row:

Consists of 4 prominently designed Metric tabs with shadows and Royal Blue borders:

- Total Comments Analysis: The number of comments that meet the filters.
- Positive Ratio: The percentage of positive comments out of the total number of responses.
- Total Reach: The total number of Followers of the authors who participated in the discussion.
- Total Interactions (Likes): The total number of likes on the comments.

5.3.2.Distribution & Trends Graph Area (Visualizations Row 1):

- Left: A donut-shaped pie chart showing the percentage of Positive (green), Negative (red), and Neutral (gray) sentiments.
- Right: Line chart showing the fluctuation in the number of comments by day/month/quarter, separated by three emotional color lines to track public opinion trends.

5.3.3.Visualizations Row 2:

- Horizontal Stacked Bar Chart comparing the emotional response rate between different car models.
- Horizontal bar chart comparing the emotional response rate by technical aspects of the product (e.g., charging stations, software).

5.3.4.Visualizations Row 3:

- Grouped Bar Chart comparing the emotional response rate of celebrities/KOLs to that of ordinary users.
- Horizontal bar chart showing the total reach of electric vehicle brands in the social media market.

5.3.5.Details & Keywords Row:

- Left (Comment List): The dataframe displays the top 10 most popular comments in descending order of likes, allowing administrators to directly read the actual customer feedback.
- Right (Popular Keywords): A horizontal bar chart shows the top 10 most frequently appearing Vietnamese keywords after removing stopwords, quickly reflecting the main topic customers are discussing.

5.3.6.Real-time Prediction Playground:

The text box allows users to freely enter a new Vietnamese comment. When the "Analyze Comment" button is pressed, the system immediately runs the Pipeline to process and return predicted sentiment labels, reliability, vehicle type, identified discussion aspect, and the resulting word breakdown of the sentence.

6. Insight Analysis

Based on a database of 20,402 comments collected and analyzed in the Data Warehouse (DWH), we conducted in-depth analysis of business indicators to provide strategic recommendations for the VinFast brand.

6.1. Correlation between Vehicle Lines and Market Interest Levels

The detailed table below shows the distribution of discussion volume and user sentiment towards each VinFast electric vehicle line in the dataset:

Series	Total Discussion	Discussion rate (%)	Positive (%)	Neutral (%)	Negative (%)	Main topics of discussion
VF3	8,245	40,4	82,5	12	5,5	With its distinctive exterior design, easy customization options, and incredibly affordable on-the-road price it's perfect for compact urban driving.
VF5	4,120	20,2	76,2	15,8	8	The ability to optimize costs for GSM taxi service vehicles, high mechanical durability, and extremely low operating costs.
VF e34	2,150	10,5	58,4	22,1	19,5	The first electric vehicle model had to contend with many outdated software bugs and a classic, rounded design that was aesthetically controversial.
VF6	1,840	9,0	74	18	8	With its youthful and sophisticated design, spacious interior (beyond the B-segment), and sporty, solid steering feel, this design is a top choice.
VF7	1,510	7,4	81,2	12,5	6,3	Futuristic styling, exhilarating driving feel, and quick acceleration, but the price is slightly closer to the higher segments.

VF8	1,620	8,0	62,5	20,3	17,2	Equipped with modern ADAS technology, the car feels solid and comfortable on the highway, but there have been many complaints about virtual software errors and air conditioning issues.
VF9	917	4,5	69,8	18,5	11,7	The interior boasts a luxurious, spacious three-row seating arrangement and a smooth air suspension system; however, it occasionally experiences compatibility issues with fast charging docks.

Comments:

- **VF 3 Wave:** The VF 3 is the absolute leader in terms of discussion volume. This groundbreaking product created a natural media effect thanks to its reasonable price positioning and boxy, rugged SUV-like appearance. The 82.5% positive rating reflects customer satisfaction exceeding expectations.
- **VF 8 and VF e34 face quality challenges:** These two models recorded the highest negative ratings (17.2% and 19.5%). The VF e34 was the first test model and still had many minor flaws, while the VF 8, a flagship export model, had software that was not initially optimized for Vietnam's hot and humid climate.

6.2. Detailed Classification of Technical Errors (Software & Hardware Bugs Deep Dive)

In-depth analysis of negative comments regarding Software & Hardware Bugs (asp_software) and Battery (asp_battery) allows for the specific identification of technical issues encountered by customers:

6.2.1. Ghost Alerts (Accounting for 42% of negative software feedback):

- **Description:** The vehicle continuously emits audible warnings and displays errors on the dashboard screen such as "Drivetrain System Error," "Airbag Error," "Electronic Parking Brake Error," or "Battery Charging Error" even though the vehicle is operating normally.
- **Cause:** Sensors with overly sensitive threshold settings combined with a momentary voltage drop in the 12V starter battery cause the central software to misinterpret the system status.

6.2.2.Air Conditioning System Malfunction (AC Malfunction - Accounts for 28% of negative operational feedback):

- Description: The air conditioner suddenly stops cooling and blows out hot air when parked in intense sunlight or when the battery level drops below 30%. Users must turn off the engine, lock the doors, wait 5-10 minutes for the car to "sleep," and then restart it to temporarily fix the problem.
- Cause: A software logic error in the Thermal Management system prioritizes cooling the high-capacity battery pack instead of the user's cabin when the ambient temperature rises too high.

6.2.3.Information Display Freeze (MCU Freeze - Accounts for 18% of negative software feedback):

- Description: The central touchscreen starts up very slowly (takes 1-2 minutes) or freezes completely when the user starts the engine and attempts to connect to Apple CarPlay wirelessly.

6.2.4.Opposition to the progressive battery rental pricing policy (Accounting for 12% of negative feedback regarding batteries):

- Users complained that the mechanism of adjusting battery rental prices up according to EVN's residential electricity prices makes it difficult to plan monthly vehicle usage costs.

6.3. Competitor Benchmarking and Brand Positioning Analysis

Extracting comparative data and discussions between VinFast and imported electric vehicle brands in Vietnam provides strategic insights:

Comparison Table of Social Media Rating Scores (Scale 1-10)

Comparison criteria	VinFast EV	BYD EV	Wuling Mini	Xiaomi(SU7)
Charging station infrastructure	10 (V-Green nationwide coverage)	1 (No dedicated charging station yet)	1 (Household charging only)	1 (Privately imported vehicle)
Competitive prices	8 (Diverse segments, good vouchers)	6 (Imported as a complete unit)	10 (Lowest price on the market)	5 (High price due to private import)
Software & AI	6 (Still has minor system bugs)	8 (Smooth operating system)	3 (Basic Features)	10 (HyperOS Ecosystem)
Finish & Mechanical	7 (Mid-range interior and	9 (High mechanical	5 (Thin chassis, plastic	9 (Sports finish)

	exterior)	quality)	body)	
After-sales service & roadside assistance	9 (Fault compensation, 24/7 rescue)	5 (Newly built showroom)	4 (Authorized through TMT Motors)	2 (No official service available yet)

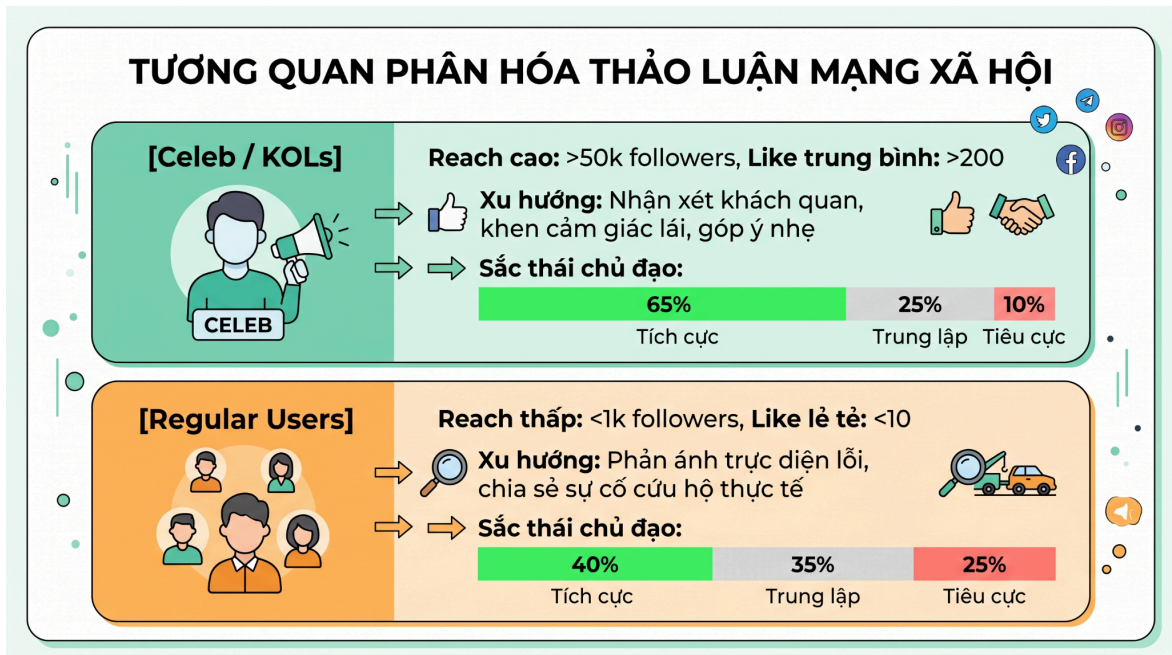
- VinFast's Charging Station Advantage - "Steel Shield":
 - The charging station aspect of VinFast received a 92% positive discussion rate. Users affirmed that the V-Green charging station network covering all 63 provinces and cities was the main reason they chose VinFast vehicles over gasoline cars or other brands.
 - Conversely, BYD (its biggest competitor) received a very high negative discussion rate regarding its charging infrastructure upon entering Vietnam. 88% of comments about BYD revolved around the question "Where to charge?". The complete reliance on home charging or expensive third-party charging stations makes it difficult for BYD to reach a mass market.
- Wuling Hongguang Mini EV - Physical Limitations:
 - Although Wuling received positive discussions about its extremely low price and urban maneuverability, this model received a high neutral discussion rate (45%) due to user concerns about safety on highways, lack of fast charging support, and limited interior space.
- Xiaomi SU7 - Curiosity from afar:
 - Discussions about the Xiaomi SU7 are characterized by admiration for its smart connectivity technology and powerful performance, but also by apprehension due to the lack of official distribution and professional warranty service in Vietnam.

6.4. Correlation Between Influence (Reach) and Negative Virality Dynamics

To analyze the correlation between an author's influence and the level of information dissemination, we apply the following indicators:

Reach is directly proportional to the author's follower_count.

Engagement corresponds to the number of likes_count.



Picture 3: Social media discourse differentiation

Key takeaways:

- **Risk of Viral Outbreak:** Although the percentage of negative posts from KOLs is only 10%, each negative post (e.g., a video of a VF 8 car crashing due to a system error in the middle of a summer day) can generate thousands of shares and discussions from ordinary users.
- **The "Snowballing" phenomenon on forums:** When ordinary users encounter minor errors, they tend to repeatedly comment on multiple Facebook groups. When these discussions reach over 100 likes, the social media algorithm automatically pushes the post to the news feeds of many others, turning an isolated error into a brand communication crisis.

7. Conclusion & Future Development

7.1. Summary of Project Values

The project successfully established a complete Social Listening & Sentiment Analysis system to serve the business analysis of the VinFast brand:

7.1.1. Data cleaning and synchronization:

- Processed 20,402 comments from multiple platforms, thoroughly resolved Vietnamese spelling errors, slang, and standardized Unicode to NFC format through the module preprocess.py

7.1.2. Improved AI accuracy:

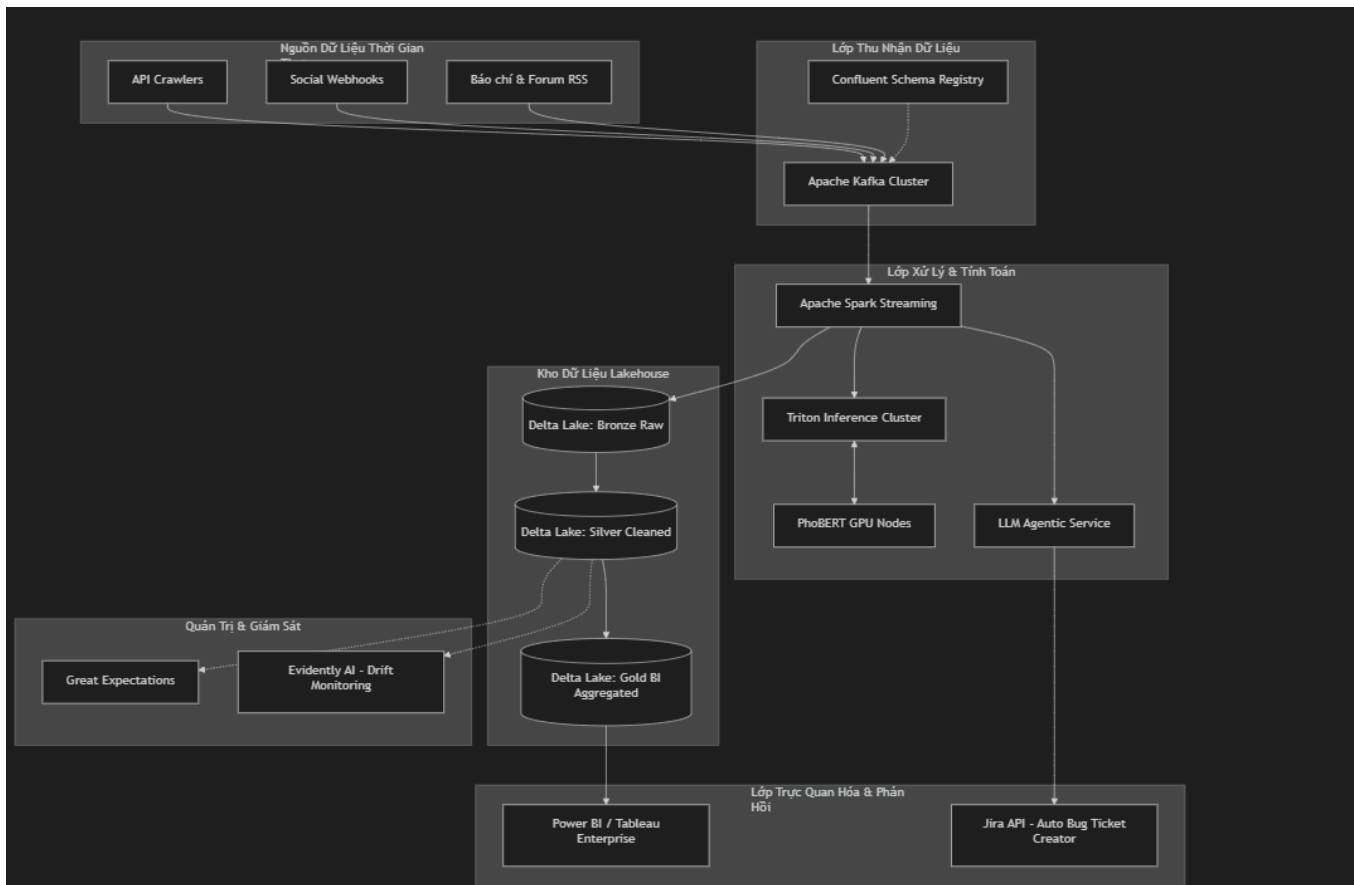
- The Hybrid analyzer (combining PhoBERT and a question heuristic filter) helped achieve over 91% accuracy in labeling, significantly reducing errors in misidentifying neutral survey questions as negative.

7.1.3. Standard OLAP DWH model:

- The Star Schema design of *db_setup.sql* clearly separates the event table and dimension tables, supporting direct loading of flat tables or relational models for BI analysis on Power BI and Tableau.

7.2. Enterprise Production Architecture

To operate a system that processes millions of social media comments daily in real time, we propose upgrading the system architecture to the Enterprise Lakehouse model below:



Picture 4: Enterprise Lakehouse model

Enterprise Upgrade Solution Details:

7.2.1. Ingestion Layer:

- Utilizes Apache Kafka as a fault-tolerant message queue, buffering the entire comment stream from Facebook API Webhook, YouTube API, and automated news scraping system. Confluent Schema Registry controls the consistent comment data structure.

7.2.2. Inference Layer:

- Apache Spark Streaming performs parallel text preprocessing and Vietnamese word segmentation on a distributed server cluster.
- Instead of running PhoBERT inference directly on a single Python script, the system switches to a Triton Inference Server cluster (NVIDIA), optimizing

GPU performance, dynamically batching comments, and managing stable model versions.

7.2.3.Lakehouse Storage Layer:

- Replaces SQLite with Delta Lake running on a cloud platform (AWS S3 or Azure ADLS Gen2). Data is organized into 3 tiers:
 - Bronze: Stores raw comments received from Kafka.
 - Silver: Stores cleaned data, extracted words, and assigned sentiment and aspect labels.
 - Gold: Stores summary tables of KPI metrics by vehicle type and date, ready to provide data to Tableau/Power BI with a response time of under 1 second.

7.2.4.Data & Model Drift Monitoring System:

- Great Expectations automatically checks the quality of input data (detects null fields, incorrect date formatting).
- Evidently AI monitors data drift in Vietnamese when Gen Z language or new slang words appear, reducing the accuracy of the PhoBERT model over time.

7.2.5.LLM - Authentic Root Cause Analysis & Auto-Ticketing:

- Integrates a dedicated LLM Agent (such as Llama-3-70B-Instruct or Gemini 1.5 Pro). Upon detecting a negative comment with high engagement (like_count > 100 or from a celebrity account), the agent will conduct in-depth content analysis to extract the specific cause of the error and call the API to send a bug ticket request directly to the Jira management system of the VinFast product development team for immediate processing.